# Prime-Shell Residue Factorization and the Deleted-Diagonal Rank Obstruction

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**Abstract**

The finite control atlas of TPC-284 shows that small changes in a prime-shell schedule can reverse the sign of a literal source attachment. This paper locates a structural reason why a tempting low-rank explanation is insufficient. For an odd prime $q$, the centered residue block factors exactly through the $q-1$ nonzero residue indicators and has rank at most $q-2$. The physical operator, however, deletes its diagonal. We prove that this deletion makes the active residue block full rank whenever every nonzero residue class occurs. The proof splits into within-class zero-sum and block-constant subspaces; a matrix-determinant-lemma factor is strictly negative. On the 20 registered prime/exponent rows, an independent modular replay further certifies that the rational kernel Schur product has full active rank. Thus the centered $q-2$ modes are a real analytic structure, but they do not by themselves furnish the literal arithmetic $L^2$ estimate. No norm bound, asymptotic cancellation, fixed-power credit, or twin-prime conclusion is claimed.

## 1 Motivation and frozen object

TPC-284 replaced the unrestricted source perturbation of TPC-283 by six named finite schedule controls and found eight baseline sign flips [1]. One possible response is to search for a small-dimensional residue model: a prime $q$ has only $q-1$ nonzero residue classes, and centering removes their constant mode. That intuition is algebraically correct before the physical diagonal convention is applied. This paper tracks the convention exactly.

Let $I$ be a finite interval of integer indices and let $q \geq 3$ be prime. Write

$$m_q(u) = \mathbf{1}_{q\nmid u}, \qquad R_q(u,a) = m_q(u)\mathbf{1}_{u\equiv a \pmod{q}}, \quad a = 1, \dots, q-1, \tag{1}$$

and let $\mathbf{1}$ denote the $(q-1)$-vector of ones. The centered residue block, including its diagonal, is

$$B_q(u,t) = m_q(u)m_q(t)\left(\mathbf{1}_{u\equiv t \pmod{q}} - \frac{1}{q-1}\right). \tag{2}$$

The literal prime-shell operator uses, for distinct $u, t$,

$$A_q(u,t) = qK_H(u-t)D_q(u,t), \qquad D_q = B_q - \operatorname{diag}(B_q), \tag{3}$$

where

$$K_H(h) = \frac{H^{2s}}{(H^2+h^2)^s}, \qquad s \in \{1,2\}. \tag{4}$$

The complete shell is a signed sum of these blocks over $Q < q \leq 2Q$ [2]. We study one block first; a full-shell estimate must still handle signs and interactions between different primes.

## 2 Exact centered factorization

**Theorem 1** (residue-mode factorization)**.** *For every odd prime $q$ and finite index set $I$,*

$$B_q = R_q P_q R_q^\top, \qquad P_q = I_{q-1} - \frac{\mathbf{1}\mathbf{1}^\top}{q-1}. \tag{5}$$

*Consequently* $\operatorname{rank}(B_q) \le q-2$*. If every nonzero residue class modulo $q$ occurs in $I$, then the rank is exactly $q-2$.*

*Proof.* The $(u,t)$ entry of $R_q R_q^\top$ is the first masked indicator in (2), while $R_q\mathbf{1} = m_q$. Substitution gives (5). The matrix $P_q$ is the orthogonal projection onto $\mathbf{1}^\perp$ and has rank $q-2$, proving the upper bound. If all classes occur, the columns of $R_q$ have disjoint nonempty supports and are linearly independent. Since $B_q = (R_q P_q)(R_q P_q)^\top$, its rank is the rank of $R_q P_q$, namely $q-2$. □

The theorem identifies the attractive residue-mode structure exactly. It is also the point at which the physical operator makes a nontrivial change.

## 3 Diagonal deletion restores full active rank

Let $J_q = \{u \in I : q \nmid u\}$ and let $n_a$ be the number of indices in $J_q$ congruent to $a$ modulo $q$. Assume $n_a > 0$ for every $a = 1, \ldots, m$, where $m = q-1$. On the active coordinates, multiply $D_q$ by $m$. Since the diagonal entry of $mB_q$ is $m-1$, we obtain

$$M_q := mD_q = mR_q R_q^\top - u_q u_q^\top - (m-1)I, \qquad u_q = R_q\mathbf{1}. \tag{6}$$

**Theorem 2** (deleted-diagonal full-rank theorem)**.** *Under the full-class condition $n_a > 0$, the active matrix $D_q$ is invertible. In particular,*

$$\operatorname{rank}(D_q) = |J_q|,$$

*even though the undeleted centered block has rank $q-2$.*

*Proof.* Decompose the active coordinate space into the direct sum of the subspaces of vectors whose sum is zero inside each residue class and the space of vectors constant on each class. On the first summand, both $R_q R_q^\top$ and $u_q u_q^\top$ vanish, so $M_q$ acts by the nonzero scalar $-(m-1)$.

For a block-constant vector with class values $c_a$, let $n = (n_1, \ldots, n_m)^\top$. The class value of $M_q c$ is

$$(mn_a - (m-1))c_a - \sum_b n_b c_b.$$

Thus the matrix on this summand is $D - \mathbf{1}n^\top$, where $D = \operatorname{diag}(d_a)$ and $d_a = mn_a - (m-1) > 0$. The matrix determinant lemma gives

$$\det(D - \mathbf{1}n^\top) = \left(\prod_a d_a\right)\left(1 - \sum_a \frac{n_a}{d_a}\right). \tag{7}$$

Because $d_a < mn_a$ for $m > 1$, each ratio $n_a/d_a$ is strictly larger than $1/m$. There are $m$ classes, so the sum in (7) is greater than one. The determinant is therefore nonzero. Both invariant summands are invertible, proving the claim. □

**Remark 3.** *The result is not a claim that diagonal deletion destroys every possible cancellation. It says something more specific and useful: a proof that uses only the $q-2$ centered residue modes cannot represent the physical off-diagonal block without paying for the deleted diagonal.*

Table 1: Finite rank audit. The modular ranks are rational-rank witnesses.

| $X$ range | $q$ values | rows | active range | rank $B_q$ | rank $D_q$, $K_H \circ D_q$ |
|---|---|---|---|---|---|
| 64 | 5,7 | 4 | 26–27 | $q-2$ | active |
| 96,128 | 7 | 4 | 41–55 | $q-2$ | active |
| 192,256 | 7,11 | 8 | 82–116 | $q-2$ | active |
| 384 | 11,13 | 4 | 175–177 | $q-2$ | active |

## 4 Finite kernel-Schur audit

We apply the theorem to the six registered tuples

$$(64, 15, 4, 4), (96, 20, 5, 4), (128, 24, 5, 4), (192, 32, 6, 5), (256, 38, 6, 5), (384, 50, 7, 5), \tag{8}$$

with $s = 1, 2$ and each prime in $Q < q \leq 2Q$. There are 20 $(X, H, Q, s, q)$ rows. For each row we check every entry of (5) exactly and record the class sizes. The following table aggregates the rank outcomes; "active" is $|J_q|$.

For the last two columns we reduce integer-scaled residue matrices and the rational kernel matrix modulo $p = 1{,}000{,}000{,}007$. Every denominator $H^2 + (u-t)^2$ is nonzero modulo $p$. A full-rank reduction gives a nonzero rational minor, so the rational matrix has full active rank. The certificate checks this independently for all 20 rows:

$$\begin{array}{c|c} \text{identity/rank test} & \text{certified rows} \\ \hline B_q = R_q P_q R_q^{\top} \text{ entrywise} & 20 \\ \operatorname{rank}(B_q) = q-2 & 20 \\ \operatorname{rank}(D_q) = |J_q| & 20 \\ \operatorname{rank}(K_H \circ D_q) = |J_q| & 20. \end{array} \tag{9}$$

The exact theorem explains the third line; the modular replay is an adversarial finite check of the implementation. The fourth line is intentionally only a finite certificate. No general theorem is asserted for the kernel Schur product, even though it is full rank on every registered row.

## 5 Consequence for the twin-prime route

The residue factorization is reusable: it separates a centered prime mode from the constant residue direction and can be inserted into a future full-shell calculation. The diagonal theorem is the corresponding obstruction. The physical matrix is not a rank-$q-2$ object after the diagonal convention, and the kernel-weighted finite blocks remain full rank. Therefore a literal arithmetic $L^2$ proof must use additional information—signed cross-prime cancellation, singular-value estimates, source orthogonality, or another structure—not merely count residue modes.

This conclusion fits the TPC-284 sign atlas. A small change in $Q$ changes which full-rank residue blocks are summed, while a change in $H$ changes their Schur weights. The observed finite sign flips are therefore compatible with the exact low-dimensional residue identity and do not justify a universal orientation claim. Conversely, full rank alone is not a negative result for cancellation; it only closes the low-rank shortcut.

## 6 Verification and conclusion

The release contains a canonical certificate, an independent checker that does not import the producer, a hostile mutation audit, and a fail-closed Bridge-B checker. The exact algebraic proof, the 20 modular rank witnesses, and the parent hash lock all pass in ordinary and optimized Python with empty standard error. The maximum claim is consequently narrow:

> Centered prime-residue modes factor exactly and have rank at most $q-2$; physical diagonal deletion restores full active rank, and the registered kernel-Schur blocks are finitely certified full rank.

The next open theorem is a signed full-shell spectral or $L^2$ bound that uses more than this rank information. Fixed-power credit remains zero, Gate B is open, and no twin-prime conclusion follows.

## References


[1] Liang Wang. A finite control atlas for literal twin-prime source attachment. TPC-284 in-repository research artifact, 2026.

[2] Liang Wang. Finite cutoff sensitivity obstruction for the literal operator. TPC-268 in-repository research artifact, 2026.